

## 5 Day Faculty Development Program on *Machine Learning Practices Used at Industry*

| Date  | Time                  | Topic                                                                                                                                                                                                                                 |
|-------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Day-1 | 10:00 am – 10:15 am   | Inaugural Session                                                                                                                                                                                                                     |
|       | 10:15 am - 01:00 pm   | <ul style="list-style-type: none"> <li>• Introduction of Artificial Intelligence and Machine Learning.</li> <li>• Evolution and technologies used in Artificial Intelligence.</li> <li>• Applications of Machine learning.</li> </ul> |
|       | 01:00 pm - 02:00 pm   | Demonstration Of Teachable Machine Learning                                                                                                                                                                                           |
|       | 02:00 pm - 03:00 pm   | Lunch Break                                                                                                                                                                                                                           |
|       | 03:00 pm - 05:00 pm   | Introduction to types of machine learning.                                                                                                                                                                                            |
|       |                       |                                                                                                                                                                                                                                       |
| Day-2 | 10:00 am - 12:00 noon | <ul style="list-style-type: none"> <li>• Introduction to Python and its Applications in AI/ML.</li> <li>• Crash Course in Python programming language</li> </ul>                                                                      |
|       | 12:00 noon - 01:00 pm | Introduction to the python libraries used in industry: pandas, numPy, etc.                                                                                                                                                            |
|       | 01:00 pm - 02:00 pm   | Lunch Break                                                                                                                                                                                                                           |
|       | 02:00 pm - 05:00 pm   | <u>Hands-on</u> with Python libraries like matplotlib, seaborn, etc                                                                                                                                                                   |
| Day-3 | 10:00 am - 12:00 noon | Introduction to Supervised Learning and its type.                                                                                                                                                                                     |
|       | 12:00 noon - 01:00 pm | <u>Hands-on</u> with Linear Regression (Regressor)                                                                                                                                                                                    |
|       | 01:00 pm - 02:00 pm   | Lunch Break                                                                                                                                                                                                                           |
|       | 02:00 pm – 05:00 pm   | <ul style="list-style-type: none"> <li>• <u>Hands-on</u> with Logistic Regression, KNN (Classifier)</li> <li>• Evaluating Supervised models: Accuracy + Confusion matrix</li> </ul>                                                   |
| Day-4 | 10:00 am - 12:00 noon | Introduction to unsupervised learning and its type.                                                                                                                                                                                   |
|       | 12:00 noon - 01:00 pm | <u>Hands on</u> with K-means                                                                                                                                                                                                          |
|       | 01:00 pm - 02:00 pm   | Lunch Break                                                                                                                                                                                                                           |
|       | 02:00 pm - 05:00 pm   | <ul style="list-style-type: none"> <li>• Saving ML Model</li> <li>• <u>Project</u> on ML with complete workflow</li> </ul>                                                                                                            |
| Day-5 | 10:00 am - 01:00 pm   | Deployment Technologies: Using Streamlit                                                                                                                                                                                              |
|       | 01:00 pm - 02:00 pm   | Lunch Break                                                                                                                                                                                                                           |
|       | 02:00 pm - 05:00 pm   | <u>Hands-on Practice</u> : Students will be given a dataset, and asked to perform prediction                                                                                                                                          |

- To get the most out of this training, student/faculty should have access to computer – personal or laboratory

